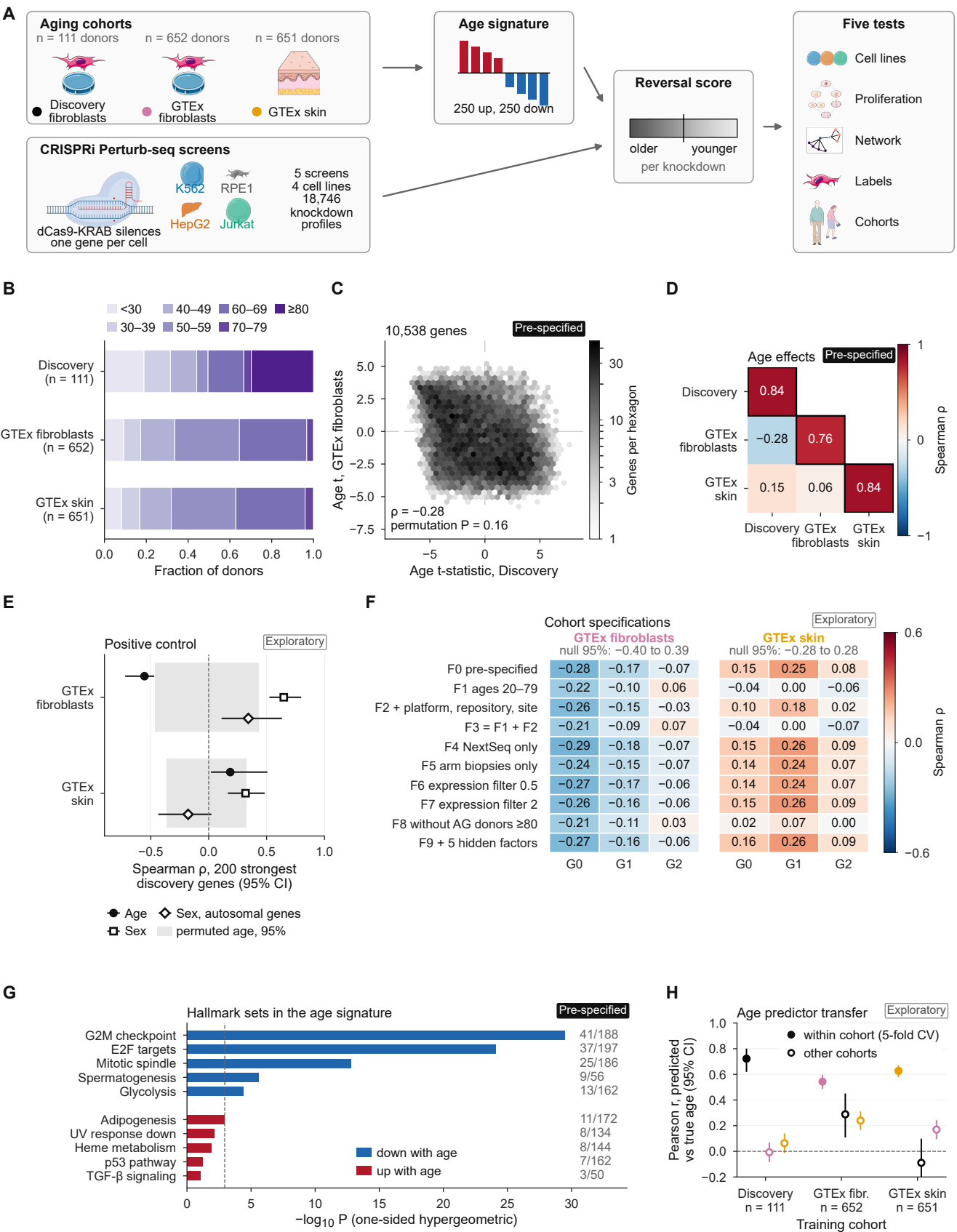

A
Aging cohorts
n = 111 donors
n = 652 donors
n = 651 donors
Discovery fibroblasts
GTEx fibroblasts
GTEx skin
Age signature
250 up, 250 down
Reversal score
older
younger
per knockdown
Five tests
Cell lines
Proliferation
Network
Labels
Cohorts
CRISPRi Perturb-seq screens
dCas9-KRAB silences one gene per cell
K562
RPE1
HepG2
Jurkat
5 screens
4 cell lines
18,746 knockdown profiles
B
<30
30–39
40–49
50–59
60–69
70–79
≥80
Discovery (n = 111)
GTEx fibroblasts (n = 652)
GTEx skin (n = 651)
Fraction of donors
C
10,538 genes
Pre-specified
Age t, GTEx fibroblasts
Age t-statistic, Discovery
Genes per hexagon
ρ = −0.28
permutation P = 0.16
D
Age effects
Pre-specified
Discovery
GTEx fibroblasts
GTEx skin
0.84
−0.28
0.76
0.15
0.06
0.84
Spearman ρ
E
Positive control
Exploratory
GTEx fibroblasts
GTEx skin
Spearman ρ, 200 strongest discovery genes (95% CI)
Age
Sex
Sex, autosomal genes
permuted age, 95%
F
Cohort specifications
Exploratory
GTEx fibroblasts
null 95%: −0.40 to 0.39
GTEx skin
null 95%: −0.28 to 0.28
F0 pre-specified −0.28 −0.17 −0.07 0.15 0.25 0.08
F1 ages 20–79 −0.22 −0.10 0.06 −0.04 0.00 −0.06
F2 + platform, repository, site −0.26 −0.15 −0.03 0.10 0.18 0.02
F3 = F1 + F2 −0.21 −0.09 0.07 −0.04 0.00 −0.07
F4 NextSeq only −0.29 −0.18 −0.07 0.15 0.26 0.09
F5 arm biopsies only −0.24 −0.15 −0.07 0.14 0.24 0.07
F6 expression filter 0.5 −0.27 −0.17 −0.06 0.14 0.24 0.07
F7 expression filter 2 −0.26 −0.16 −0.06 0.15 0.26 0.09
F8 without AG donors ≥80 −0.21 −0.11 0.03 0.02 0.07 0.00
F9 + 5 hidden factors −0.27 −0.16 −0.06 0.16 0.26 0.09
G0 G1 G2 G0 G1 G2
Spearman ρ
G
Hallmark sets in the age signature
Pre-specified
G2M checkpoint 41/188
E2F targets 37/197
Mitotic spindle 25/186
Spermatogenesis 9/56
Glycolysis 13/162
Adipogenesis 11/172
UV response down 8/134
Heme metabolism 8/144
p53 pathway 7/162
TGF-β signaling 3/50
down with age
up with age
−log10 P (one-sided hypergeometric)
H
Age predictor transfer
Exploratory
within cohort (5-fold CV)
other cohorts
Pearson r, predicted vs true age (95% CI)
Discovery n = 111
GTEx fibr. n = 652
GTEx skin n = 651
Training cohort